

Unit-I

Numerical techniques

**Numerical solution of algebraic and transcendental
equations:**

False position methods



॥वसुधैव कुटुम्बकम्॥

Dr. Ram Kishun Lodhi

Department of Applied Science

Symbiosis Institute of Technology Pune

False Position Methods or Regula-Falsi Method

This is the oldest method of finding the real root of an equation $f(x) = 0$ and closely resembles the bisection method.

- Identify two points a and b such that $f(a)$ and $f(b)$ are of opposite signs.
- Hence the root lies between a and b .
- The equation of line joining the two points $(a, f(a))$ and $(b, f(b))$ is

$$\frac{y - f(a)}{f(b) - f(a)} = \frac{x - a}{b - a} \quad (1)$$

- The solution indicate some point on x-axis, then y become zero i.e.

$y=0$. From eq (i), we get

$$\frac{0 - f(a)}{f(b) - f(a)} = \frac{x - a}{b - a}$$

$$\Rightarrow x = \frac{a f(b) - b f(a)}{f(b) - f(a)}.$$

This gives the first approximation to the root for $f(x) = 0$.

In general: $\Rightarrow x_{n+2} = \frac{x_n f(x_{n+1}) - x_{n+1} f(x_n)}{f(x_{n+1}) - f(x_n)}.$

- We continue this until the two successive approximation is approximately equal.

Example: Find the root of the equation $\cos x = xe^x$ using the regula-falsi method correct to four decimal places.

Solution: Let $f(x) = \cos x - xe^x = 0$

So that $f(0) = 1$, $f(1) = \cos 1 - e = -2.17798$.

i.e. the root lies between 0 and 1.

Taking $a = 0$, $b = 1$, $f(0) = 1$ and $f(1) = -2.17798$ in the regula-falsi

method, we get

$$x = \frac{a f(b) - b f(a)}{f(b) - f(a)} \quad (1)$$

First Approximation:

$$x_1 = \frac{0 \times -2.17798 - 1 \times 1}{-2.17798 - 1} = 0.31467.$$

Now $f(0.31467) = 0.51987$ i.e., the root lies between 0.31467 and 1.

Second Approximation

Taking $a = 0.31467$, $b = 1$, $f(0.31467) = 0.51987$ and $f(1) = -2.17798$

From eq. (1), we get

$$x_2 = \frac{0.31467 \times -2.17798 - 1 \times 0.51987}{-2.17798 - 0.51987} = 0.44673.$$

Now $f(0.44673) = 0.20356$ i.e., the root lies between 0.44673 and 1.

Repeating this process, the successive approximations are

$$x_3 = 0.49402.$$

$$x_4 = 0.50995.$$

$$x_5 = 0.51520.$$

$$x_6 = 0.51692.$$

$$x_7 = 0.51748.$$

$$x_8 = 0.51767.$$

$$x_9 = 0.51775.$$

Hence the root is 0.5177 correct to 4 decimal places.

Q.1: Use the method of false position to find the fourth root of 32 correct to three decimal places.

Answer: 2.378.

Q.2: Find a real root of the equation $x \log_{10} x = 1.2$ by regula-falsi method correct to four decimal places.

Answer: 2.7406.

Thank You